SIEMENS

Installation Instructions
Models CC-5 / CC-2

System Cardcage CC-5 (500-633037 / S24235-B123-A1)
System Cardcage CC-2 (500-633440 / S24235-B122-A1)

INTRODUCTION

The SIEMENS Model CC-5 Cardcage (see Figure 1) has five card slots and fits in two positions on the Mounting Plate. The SIEMENS Model CC-2 Cardcage (see Figure 2) has two card slots and fits in one position on the Mounting Plate. The CC-5 and CC-2 are used for mounting all system circuit card assemblies. They each have two removable terminal block connectors at the top and one removable terminal block connector at the bottom of each card slot. All field-wiring connections are made directly to the top of the cardcage while internal power is connected to the bottom of the cardcage. The two removable terminal block connectors at the top of each slot are arranged in two rows. The lower row is labeled 1 through 8 and the upper row is labeled 9 through 16. These removable terminal block connectors are used for connecting power limited external equipment. The removable terminal block connector located at the bottom of each slot is labeled 17 through 24 and is used for connecting internal power.

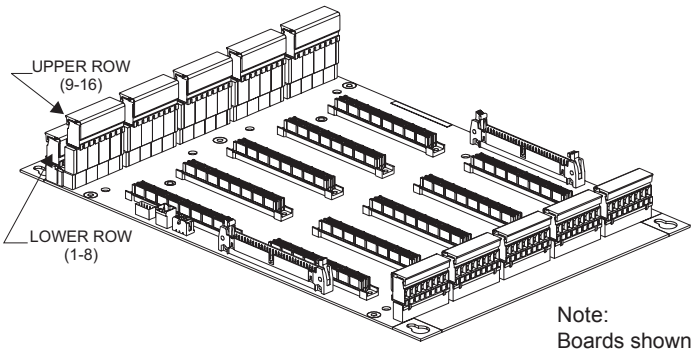


Figure 1
CC-5 Card Cage

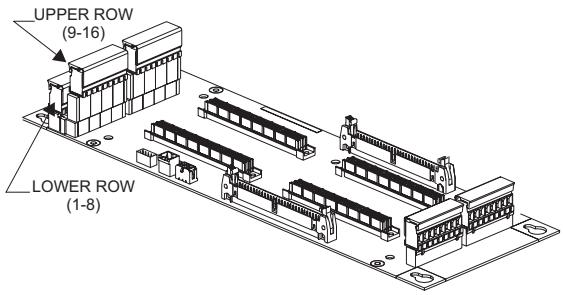


Figure 2
CC-2 Card Cage

Note:
Boards shown without card guides installed.

OPERATION

The CC-5 and CC-2 provide a central point for mounting circuit card assemblies. If a card requires power, it is applied through the removable terminal block at the bottom of the cardcage. All external devices for the card are connected to the card through the two removable terminal blocks at the top of the cardcage. The circuit card assemblies are connected to the HNET and CAN bus through the connectors on the cardcage.

P/N 315-033035-3
A24205-A334-B815

PRE-INSTALLATION

The CC-5 and CC-2 are shipped with the card guides removed. The card guides are keyed to assist in installation. They can be installed either before or after the cardcage is installed in the enclosure. For ease of installation, the card guides have been designed to be installed after the field wiring has been completed.

Card Guide Installation

(Figure 3)

CAUTION

The upper and lower card guides are keyed differently and are not interchangeable.

Ensure that the card guide keys fit easily into the holes in the motherboard to prevent damage.

1. The upper card guide has a label on each slot that can be used to identify the module installed in the slot. If desired, complete the label before inserting the upper card guide.
2. Insert the upper card guide and secure with the captive screws.
3. Insert the lower card guide and secure with the captive screws.

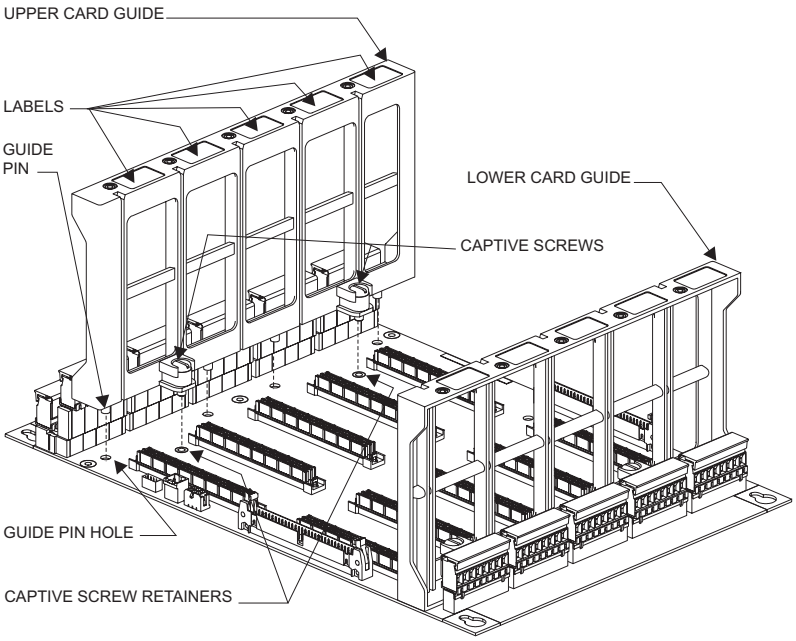


Figure 3
Card Guide Installation (CC-5 Shown)

INSTALLATION

WARNING

Disconnect BATTERY and AC prior to working on equipment.

The CC-5 and CC-2 mount on the Mounting Plate. The Mounting Plate may be located either “IN” or “OUT” of the enclosure to perform this inst allation procedure. If the Mounting Plate is located “IN” the enclosure you will have to gain access to it

by opening the enclosure Inner and Outer doors. If the Mounting Plate is located outside of the enclosure, place it in front of you so that the word “T OP” is at the top and away from you.

Each Mounting Plate can hold a maximum of two CC-5 cardcages. There are three possible locations for mounting the CC-5 on the Mounting Plate.

1. Flush left on the Mounting Plate in positions 1-2.
2. In the center of the Mounting Plate in positions 2-3.
3. Flush right on the Mounting Plate in positions 3-4.

The CC-2 is most effectively used to add two more card slots in a row that already has one CC-5 and another 1-position module (i.e., PSC-12). It is best installed in a position adjacent to the CC-5.

In all of the possible mounting locations, the top and bottom of the CC-5 and CC-2 will be flush with the top and bottom of the Mounting Plate.

When only one CC-5 cardcage is to be installed, it is normally installed on the left side of the Mounting Plate. There is a top and bottom to the Mounting Plate. The top is where the word “TOP” is stamped on the Mounting Plate. When the cardcage is mounted correctly on the Mounting Plate it will be flush on the top, bottom, and left with the Mounting Plate and the two screw holes in the center of the cardcage motherboard will be over threaded posts.

1. Install four 10-32 screws (M5) in the threaded posts at the top left of Position 1, top right of Position 2, bottom left of Position 1, and the bottom right of Position 2 on the Mounting Plate. Screw each of the 10-32 screws (M5) into the threaded posts 5-6 turns. (Refer to Figure 3.)
2. Place the cardcage over the four screws on the Mounting Plate and slide it down or towards you to rest on the four screws. (Refer to Figure 4.) When the cardcage is in the correct position it will be flush with the top, bottom, and left side of the Mounting Plate.
3. Install the two 10-32 screws (M5) in the center of the cardcage motherboard near the left and right edge.
4. Tighten the six screws.

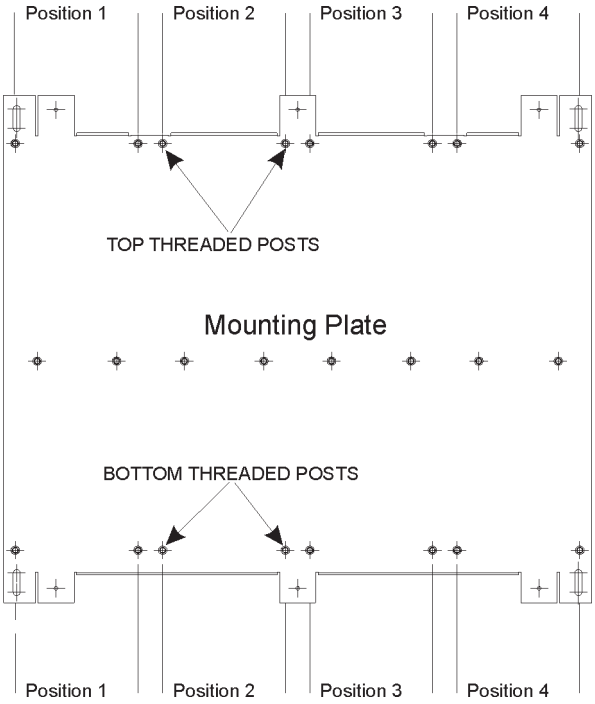


Figure 4
Location Of The CC-5 and CC-2 On The Mounting Plate

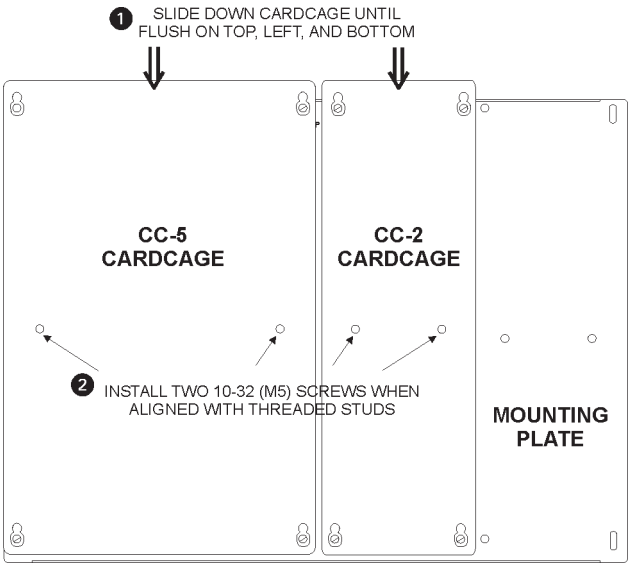


Figure 5
Mounting The CC-5 and CC-2 On Mounting Plate

WIRING

At the top and bottom of the CC-5 and CC-2 and adjacent to each card slot are removable terminal blocks. The removable terminal blocks are color-coded to each slot and have a white cover over the top. The wiring to the two terminal block connectors at the top of the CC-5 and CC-2 is power limited. The wiring to the terminal block connector at the bottom of the CC-5 and CC-2 is non-power limited.

To Connect External Wiring To The Card Slots:

(Figure 6)

1. On the removable terminal block, open the white cover .
2. Insert the wire to be connected in the desired opening in the side of the removable terminal block.
3. Tighten the appropriate locking screw.

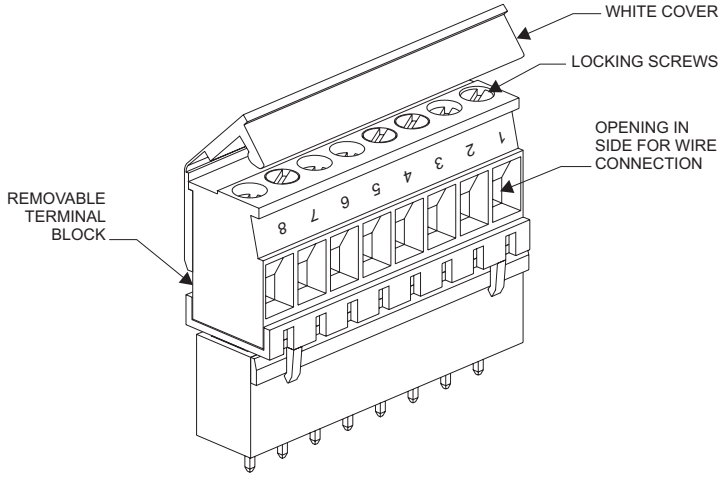


Figure 6
Removable Terminal Block On The CC-5 and CC-2

To Connect The Data Bus:

The Data bus connectors on the cardcage motherboard are P1 and P2. (Refer to Figure 7.) They are 60 pin, keyed, male connectors with connector extractors.

1. Open the connector ejectors out of the vertical position and away from the connector body.
2. Insert the female cable connector into the male connector (P1 or P2) on the cardcage motherboard. Gently press the cable connector into the motherboard connector. As the cable connector is pressed in, the connector ejectors will begin to move to the vertical position. When the cable connector is fully seated the connector ejectors will be vertical.

NOTE

Use the 60 pin bus cable - long, P/N 500-633997 / C24235-A1-K9 when the PSC-12C/ PSX-12C is located on a different Mounting Plate

To Connect The CAN Bus:

The CAN bus (P3) is a six pin connector located just above P1 on the cardcage motherboard. (Refer to Figure 7.) It is used to supply CAN communications and power to the Switch Control Modules (SCM-8S), LED, Fan Control Modules (FCM-6S), Live Voice Microphone Module (LVM) and Firefighters Master Telephone (FMT). The connector is keyed and requires no special instructions for cable connection. Use cable CCL, P/N 500-634214 / C24235-A1-K6 to connect from CAN bus (P3) on CC-5 and CC-2 to the desired module. CAN power must be supplied to E11 (CC-5) or E5 (CC-2) Refer to Figure 8 for wiring details.

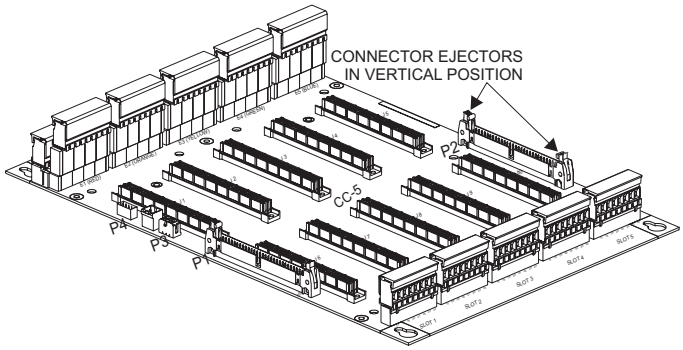


Figure 7
Location Of Connectors (CC-5 Shown)

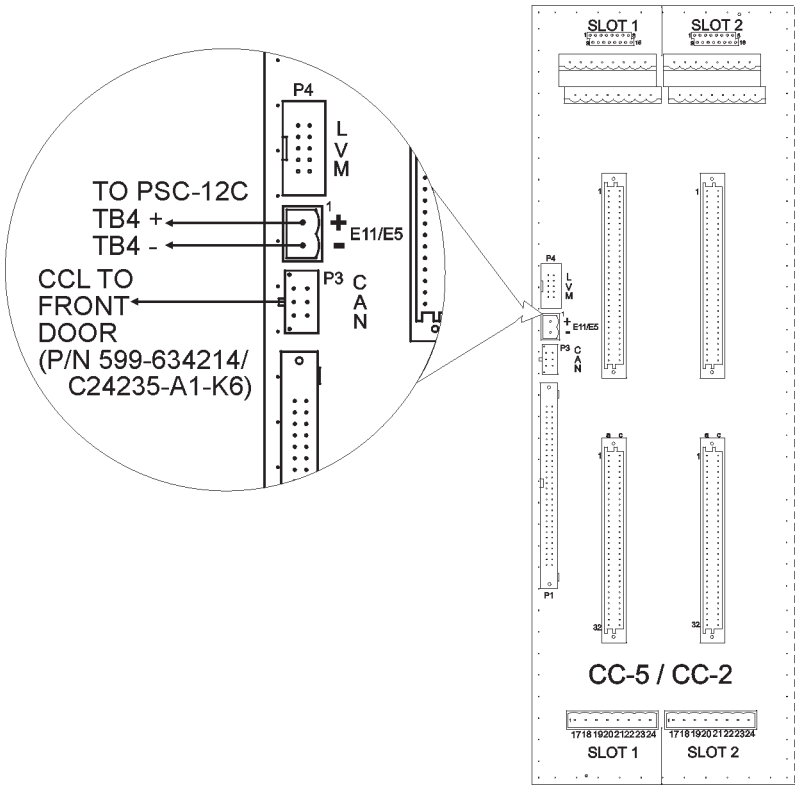


Figure 8
Connecting Power For CAN Bus

SPECIFICATIONS

Physical Specifications: CC-5 10.58”W x 14.25” H (26.87cm x 36.20cm)

CC-2 4.78”W x 14.25”H (12.14cm x 36.20cm)

Electrical Ratings: The PSC-12C supplies the power to the two 60-pin Data bus connectors (P1 and P2) in the CC-5 and CC-2. The total rating on P1 and P2 for all CC-5s in any given enclosure is as follows:

24 VDC, 2 Amps max
6.2 VDC, 2 Amps max

See PSC-12C Installation Instructions, P/N 315-033060 / A24205-A334-B819 and the PSX-12C Installation Instructions, P/N 315-034050 / A24205-A334-B818, for complete power supply loading calculations.

For CE applications in Cerberus E100 systems refer to
Installation Instruction A24205-A334-B844 (English) or A24205-A334-A844 (German).

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